

My Tasco AI Hackathon API Documentation

Reusable API and mock SDK pack for an AI-powered enterprise workforce assistant: staff directory search, AI usage analytics, organization graph, attendance, requests & approvals, payroll insights, news feed, and notifications.

Version	2026-06-26
Application code	MYTASCO (sent as X-App-Code header)
Mock base URL	http://localhost:8788
Recommended facade	https://hackathon.example.com/mytasco/v1
Current COP gateway (dev)	https://dev-api-swm.dnpwater.vn/cop/
Production base URLs	Resolved at runtime via Firebase Remote Config

COP Microservice Prefixes

My Tasco (Flutter package `dnp_water`) talks to a set of COP microservices behind a shared gateway. Each service is bound to its own named HTTP client with an independent base URL. Endpoints in this document are written relative to their service prefix.

Service	Prefix (dev)	Client	Used by
auth	/cop/auth/auth/	AuthenticatorClient	login, token refresh
iam	/cop/iam/	CoreClient	user info, face verify
sys	/cop/sys/	SysClient	staff search, global lists
hrm	/cop/hrm/	HrmClient	org, attendance, requests, salary, news
aiwsp	/cop/aiwsp/	AiwspClient	AI usage analytics
noti	/cop/noti/	NotiClient	notifications
crm / fsm / asm / wdn	/cop/crm fsm asm wdn /	Crm/Fms/Asset/WdnClient	customers, files, assets, water network

Overview

This document defines the API surface that hackathon teams should target when building AI assistants, copilots, semantic search, or analytics on top of My Tasco - the DNP Water internal workforce super-app. The goal is pragmatic reuse: a solution should be callable from the current Flutter app through a thin client adapter and should return stable, app-compatible DTOs.

Current App Integration Points

The current app already centralizes its service boundaries here:

- Network layer / token refresh / error handling: `BaseApi` (`lib/common/api/base_api.dart`)
- Per-service HTTP clients & base URLs: `DioClient` subclasses (`lib/common/api/dio_client.dart`)
- Endpoint registry (single source of truth for paths): `RouteApi` (`lib/common/api/route_api.dart`)
- Environment base URLs (dev hardcoded / prod Remote Config): `AppConfig` (`lib/app/config/app_config.dart`)
- App-compatible models, e.g. `StaffModel` (`lib/module/ticket/model/staff_model.dart`) and `AiUsageOverviewGeneralSdo` (`lib/module/ai_overview/models`)

A hackathon solution can either implement the normalized DTOs below directly, or ship a thin adapter that maps an existing backend response into them.

Authentication

Production endpoints require a Bearer access token. The app obtains it at login and refreshes it transparently: on a 401 with an expired-token code, BaseApi calls auth/refresh-token once (guarded by a shared completer to avoid duplicate refreshes) and retries the original request. Hackathon mock APIs may accept requests with or without authentication.

Do not hardcode credentials in mobile code. Keep the SDK constructor configurable with baseUrl, a bearer-token provider, and the app-code header.

Recommended common headers

```
Authorization: Bearer <access_token>
X-App-Code: MYTASCO
X-App-Version: 1.3.3
X-Device-Info: <base64({"deviceId":..., "deviceName":..., "deviceType": "ios|android"})>
X-Request-Id: 3b1d0fb1-7a45-4d2c-9f57-9c0fd85a9b9d # echoed back as x-request-id
Content-Type: application/json
X-Locale: vi-VN
X-Timezone: Asia/Ho_Chi_Minh
```

Response Envelope

Every COP response wraps its payload in a body field. Success is signalled by status: "success" (or message: "SUCCESS"). The app's extractData helper returns body directly to callers; SDKs should do the same.

```
{
  "status": "success",
  "message": "SUCCESS",
  "body": { /* endpoint-specific payload */ },
  "requestId": "3b1d0fb1-7a45-4d2c-9f57-9c0fd85a9b9d"
}
```

Common DTOs

StaffSummary

Returned by staff search / directory endpoints (maps to StaffModel).

```
{
  "staffId": 10472,
  "staffCode": "NS-10472",
  "staffName": "Nguyen Van A",
  "title": "Field Technician",
  "email": "a.nguyen@dnwater.vn",
  "phoneNumber": "0901234567",
  "status": 1,
  "provinceName": "Ha Noi",
  "listOrgUnit": [
    { "orgUnitId": 32, "orgUnitCode": "PB-KT", "orgUnitName": "Phong Ky Thuat" }
  ]
}
```

Paged request / response

List endpoints (e.g. staff search) use an example filter object plus a pageInfo block on the request, and return result with a pageInfo echo.

```
// request body
{
  "example": {
    "orgUnitId": 32,
    "status": 1,
    "keyword": "nguyen"
  },
  "pageInfo": {
    "pageSize": 400,
    "currentPage": 0
  }
}

// response body
{
  "result": [ /* StaffSummary[] */ ],
  "pageInfo": {
    "pageSize": 400,
    "currentPage": 0,
    "totalRecord": 27
  }
}
```

ErrorResponse

```
{
  "status": "error",
  "code": "invalid_request",
  "message": "keyword is required",
  "technicalMessage": "validation failed on field 'keyword'",
  "requestId": "3b1d0fb1-7a45-4d2c-9f57-9c0fd85a9b9d"
}
```

Common error codes

HTTP	Code	Meaning
400	invalid_request	Missing or invalid parameter / body
401	unauthorized	Missing or expired token (app auto-refreshes once, then retries)
403	forbidden	Authenticated but not allowed for this resource / org scope
404	not_found	Resource not found
408	timeout	Upstream / service timeout
429	rate_limited	Too many requests
500	internal_error	Unexpected service error
503	service_unavailable	Upstream unavailable

1. Staff Directory Search API

Search the employee directory by name, code, phone, or email - the primary entity-resolution endpoint for an assistant ('who is...', 'find...', '@mention').

Endpoint

POST {sys}/staff/search

Aliases

POST {sys}/staff/quick-search # lighter projection for autocomplete

Request example

```
POST {sys}/staff/search
Authorization: Bearer <token>
Content-Type: application/json

{
  "example": { "orgUnitId": 32, "status": 1, "keyword": "nguyen" },
  "pageInfo": { "pageSize": 20, "currentPage": 0 }
}
```

Supported parameters

Name	Type	Required	Description
example.keyword	string	no	Free-text match on name / code / phone / email
example.orgUnitId	number	no	Restrict to an organization unit
example.status	integer	no	Employment status filter (1 = active)
example.sortSpecialId	array	no	Staff IDs to pin to the top of results
pageInfo.pageSize	integer	no	Page size, default app value 400
pageInfo.currentPage	integer	no	Zero-based page index

Response example

```
{
  "result": [
    {
      "staffId": 10472, "staffCode": "NS-10472",
      "staffName": "Nguyen Van A", "title": "Field Technician",
      "email": "a.nguyen@dnwater.vn", "phoneNumber": "0901234567",
      "listOrgUnit": [ { "orgUnitId": 32, "orgUnitName": "Phong Ky Thuat" } ]
    }
  ],
  "pageInfo": { "pageSize": 20, "currentPage": 0, "totalRecord": 27 }
}
```

Error codes: invalid_request, unauthorized, forbidden, rate_limited, internal_error

2. AI Usage Overview API

Aggregated AI-assistant usage and cost analytics across the organization - KPIs, time-series trends, and ranking by top-level org unit. Powers dashboards and 'how are we using AI?' answers.

Endpoint

GET {aiwsp}/ai-usage-overview/general

Request example

GET {aiwsp}/ai-usage-overview/general?organizationLv1s=1,2&departmentIds=32&usageTimeFrom=2026-06-01&usageTimeTo=2026-06-25
Authorization: Bearer <token>

Supported parameters

Name	Type	Required	Description
organizationLv1s	string	no	Comma list of top-level organization IDs
organizationIds	string	no	Comma list of organization IDs
departmentIds	string	no	Comma list of department IDs
usageTimeFrom	string	no	Start date (YYYY-MM-DD)
usageTimeTo	string	no	End date (YYYY-MM-DD)

Response example

```
{
  "general": {
    "totalCost": 184.52, "percentageChangeTotalCost": 12.4,
    "totalQuestion": 8421, "percentageChangeTotalQuestion": 9.1,
    "totalUser": 312, "totalToken": 5120340,
    "percentError": 1.8, "avgMillis": 2400
  },
  "usageTrends": [ { "usageTime": "2026-06-25", "totalQuestion": 410 } ],
  "costTrends": [ { "usageDate": "2026-06-25", "totalCost": 9.12 } ],
  "rankByOrganizationLv1": [ { "organizationName": "NS3", "totalQuestion": 3120 } ]
}
```

Error codes: invalid_request, unauthorized, forbidden, service_unavailable, internal_error

3. Organization Graph API

The organization / department tree used to scope every other query and to resolve reporting lines.

Endpoint

```
GET {hrm}/organization/tree
```

Aliases

```
GET {hrm}/staff-org/by-current-staff # org units of the current staff
GET {sys}/orgUnit/initTree         # management department tree
```

Request example

```
GET {hrm}/organization/tree
Authorization: Bearer <token>
```

Supported parameters

Name	Type	Required	Description
organizationId	number	no	Subtree root; omit for full tree
depth	integer	no	Max levels to expand

Response example

```
{
  "result": [
    {
      "organizationId": 1, "organizationName": "DNP Water",
      "organizationCode": "DNP", "parentId": null,
      "children": [
        { "organizationId": 32, "organizationName": "Phong Ky Thuat", "parentId": 1 }
      ]
    }
  ]
}
```

Error codes: unauthorized, forbidden, not_found, service_unavailable, internal_error

4. Attendance API

Per-staff attendance entries and monthly summaries - the data behind 'did I check in today?' and 'how many late days this month?' assistant queries.

Endpoint

```
GET {hrm}/attendance/by-staff
```

Aliases

```
GET {hrm}/attendance/summary-by-staff    # aggregated counters
POST {hrm}/attendance/search             # filtered search
GET {hrm}/attendance-entry/self-list?date=YYYY-MM-DD
```

Request example

```
GET {hrm}/attendance/by-staff?staffId=10472&fromDate=2026-06-01&toDate=2026-06-26
Authorization: Bearer <token>
```

Supported parameters

Name	Type	Required	Description
staffId	number	no	Target staff; defaults to current user
fromDate	string	yes	Range start (YYYY-MM-DD)
toDate	string	yes	Range end (YYYY-MM-DD)
date	string	no	Single day for the self-list variant

Response example

```
{
  "result": [
    {
      "date": "2026-06-26", "checkIn": "08:02", "checkOut": "17:35",
      "workingHours": 8.5, "status": "on_time", "location": "HQ Ha Noi"
    }
  ],
  "summary": { "workingDays": 21, "lateDays": 2, "absentDays": 0 }
}
```

Error codes: invalid_request, unauthorized, forbidden, not_found, internal_error

5. Requests & Approvals API

Create, search, and act on workforce requests (leave, overtime, etc.). Approval actions let an assistant draft and route requests on the user's behalf.

Endpoint

POST {hrm}/request/search

Aliases

POST {hrm}/request/create	GET {hrm}/request/self?id=...
POST {hrm}/request/approve	POST {hrm}/request/reject
POST {hrm}/request/cancel	GET {hrm}/request/count

Request example

```
POST {hrm}/request/search
Content-Type: application/json

{
  "example": { "status": "PENDING", "requestType": "LEAVE" },
  "pageInfo": { "pageSize": 20, "currentPage": 0 }
}
```

Supported parameters

Name	Type	Required	Description
example.status	string	no	PENDING, APPROVED, REJECTED, CANCELLED
example.requestType	string	no	Request category (e.g. LEAVE, OVERTIME)
example.fromDate / toDate	string	no	Date range filter
pageInfo.pageSize / currentPage	integer	no	Pagination
id (action body)	number	yes*	Required by approve / reject / cancel / self

Response example

```
{
  "result": [
    {
      "id": 5012, "requestType": "LEAVE", "status": "PENDING",
      "staffId": 10472, "fromDate": "2026-07-01", "toDate": "2026-07-03",
      "reason": "Annual leave", "currentApprover": "NS-10001"
    }
  ],
  "pageInfo": { "totalRecord": 4 }
}
```

Error codes: invalid_request, unauthorized, forbidden, not_found, internal_error

6. Payroll & Income API

Self-service salary, payslip, and income-trend data for the signed-in employee. Sensitive: access is OTP-gated in the app, so SDKs must support a step-up token / OTP flow.

Endpoint

```
GET {hrm}/salary-staff-payslip/self-by-month
```

Aliases

```
GET {hrm}/salary-staff/self-by-month      # income breakdown
GET {hrm}/salary-staff/stats-month       # monthly chart series
GET {hrm}/salary-staff-payslip/self-by-year # yearly totals
```

Request example

```
GET {hrm}/salary-staff-payslip/self-by-month?month=6&year=2026
Authorization: Bearer <token>
```

Supported parameters

Name	Type	Required	Description
month	integer	yes	Target month (1-12)
year	integer	yes	Target year
staffOrgId	number	no	Specific staff-org assignment
otp / otpTransactionId	string	no	Step-up verification for sensitive data

Response example

```
{
  "month": 6, "year": 2026,
  "grossIncome": 18500000, "netIncome": 16320000,
  "items": [
    { "code": "BASE", "name": "Base salary", "amount": 15000000 },
    { "code": "OT", "name": "Overtime", "amount": 1200000 },
    { "code": "INS", "name": "Insurance", "amount": -1080000 }
  ]
}
```

Error codes: invalid_request, unauthorized, forbidden, not_found, internal_error

7. News & Feed API

Internal news / announcements feed with categories, reactions, and a 'latest' shortcut - a summarization-friendly content source for the assistant.

Endpoint

```
GET {hrm}/news-article/search
```

Aliases

```
GET {hrm}/news-article/latest      GET {hrm}/news-article/self?id=...
GET {hrm}/news-category/search    POST {hrm}/news-article/react
GET {hrm}/banner/visible          # active home banners
```

Request example

```
GET {hrm}/news-article/search?keyword=safety&categoryId=4&page=0&pageSize=10
Authorization: Bearer <token>
```

Supported parameters

Name	Type	Required	Description
keyword	string	no	Full-text query over title / content
categoryId	number	no	Filter by news category
status	string	no	PUBLISHED, DRAFT, PENDING, etc.
page / pageSize	integer	no	Pagination (page is zero-based)

Response example

```
{
  "result": [
    {
      "id": 880, "title": "Safety guidelines Q3",
      "categoryName": "HSE", "status": "PUBLISHED",
      "publishedAt": "2026-06-20T03:00:00Z", "reactionCount": 42,
      "summary": "Updated field-work safety rules effective July."
    }
  ],
  "pageInfo": { "page": 0, "pageSize": 10, "totalRecord": 13 }
}
```

Error codes: invalid_request, unauthorized, forbidden, rate_limited, internal_error

8. Notifications API

Per-user notification inbox with unread counts and read receipts - the channel an assistant uses to surface 'what needs my attention?'.

Endpoint

```
GET {noti}/noti-app/search
```

Aliases

```
GET {noti}/noti-app/count-unread
```

```
POST {noti}/noti-app/mark-read
```

Request example

```
GET {noti}/noti-app/search?currentPage=0&pageSize=20&isRead=false
Authorization: Bearer <token>
```

Supported parameters

Name	Type	Required	Description
currentPage	integer	no	Zero-based page index
pageSize	integer	no	Page size
isRead	boolean	no	Filter by read state; omit for all
id (mark-read body)	number	yes*	Notification ID for mark-read

Response example

```
{
  "result": [
    {
      "id": 99312, "title": "Request approved",
      "body": "Your leave request #5012 was approved.",
      "type": "REQUEST", "isRead": false,
      "createdAt": "2026-06-26T01:10:00Z", "refId": 5012
    }
  ],
  "pageInfo": { "currentPage": 0, "pageSize": 20, "totalRecord": 5 }
}
```

Error codes: invalid_request, unauthorized, forbidden, rate_limited, internal_error

Mock Server

Run a deterministic mock that mirrors the COP body envelope and the endpoints above:

```
node docs/hackathon/mock_mytasco_server.js # listens on :8788

# Health check
curl "http://localhost:8788/health"

# Staff search (short alias)
curl -X POST "http://localhost:8788/staff/search" \
  -H "Content-Type: application/json" \
  -d '{"example":{"keyword":"nguyen"},"pageInfo":{"pageSize":5,"currentPage":0}}'
```

Expected shape

```
{
  "status": "success",
  "body": {
    "result": [ { "staffId": 10472, "staffName": "Nguyen Van A" } ],
    "pageInfo": { "pageSize": 5, "currentPage": 0, "totalRecord": 1 }
  }
}
```

Submission Expectations For Hackathon Teams

Teams should submit:

- API endpoint or deployable service.
- OpenAPI / Swagger spec or equivalent contract.
- SDK / client adapter for Flutter / Dart, plus REST examples.
- Mock / staging endpoint and deterministic sample data.
- Notes on ranking / enrichment logic, latency, fallback behavior, and data provenance.

Compatibility requirements

- Preserve the COP response envelope: payload under body, status: "success" on success, ErrorResponse on failure.
- Return stable IDs (staffId, organizationId, request / article / notification IDs).
- Preserve Vietnamese text and diacritics in real services (UTF-8 end to end).
- Support a configurable base URL, bearer-token provider, and the X-App-Code: MYTASCO header.
- Honor org-scope and the OTP step-up flow for sensitive data (payroll).
- Avoid app-specific UI dependencies in the service / SDK layer.
- Provide deterministic mock data for tests and demos.